Lived Experience as Critical Data

An Inclusive Research Method

Jen Strickland
Lead Designer, Technology Platform Director
Maryland Department of Information Technology
previously with The MITRE Corporation

Approved for Public Release; Distribution Unlimited. Public Release Case Number 25-1602. The author's affiliation with The MITRE Corporation is provided for identification purposes only and is not intended to convey or imply MITRE's concurrence with, or support for, the positions, opinions, or viewpoints expressed by the author. ©2025 The MITRE Corporation. ALL RIGHTS RESERVED.

Hi! I'm Jen Strickland.

Designer, Developer, Researcher, Strategist, Policy-wonk-in-training in D.C.

- These days I'm a Lead Designer, Technology Platform Director, at the Maryland Department of Information Technology.
- I'm earning a master's in public administration at the Maxwell School of Citizenship & Public Affairs at Syracuse University.
- I'm an Invited Expert with the Worldwide Web Consortium (W3C) Accessibility Guidelines Working Group. I chair a community group and am active on other working groups. If you're looking to participate with the W3C, please reach out and I'll help connect you.
- I also volunteer with the U.S. Digital Response and Tech Talent Project.

Maryland DEPARTMENT OF INFORMATION TECHNOLOGY

Syracuse University Maxwell School of Citizenship & Public Affairs

US DIGITAL RESPONSE.org

WORLD WIDE WEB CONSORTIUM

MITRE SOLVING PROBLEMS FOR A SAFER WORLD™

© 2025 THE MITRE CORPORATION. ALL RIGHTS RESERVED.

Acknowledgement

How can we be in good relationship with Indigenous peoples, with non-human beings, with the land and water?

Currently we are on ancestral lands of the:

Tongva

(Gabrieleño / Fernandeño)

Photo by Laurie Avocado, Creative Commons

Acjachemen

(called Juaneño by the Spanish colonizers)

License: Public domain.
Credit: UC Irvine.

Questions

- How can rigorous qualitative methods be adapted to public policy contexts to effectively broaden participation in STEM institutions and organizations?
- What if qualitative research methods are not included in public policy processes?
- What qualitative methodologies can support broadening participation efforts?

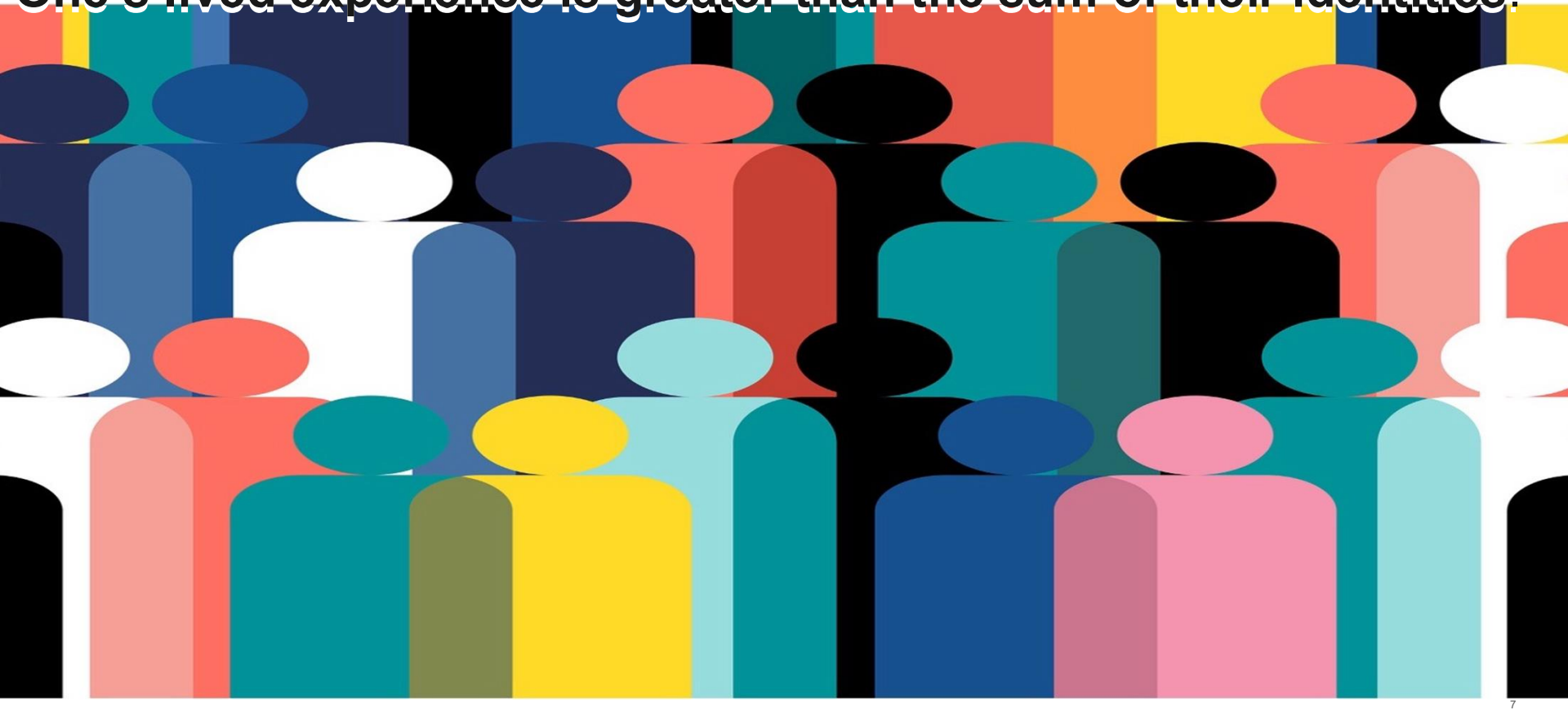
Quote from an interview participant

“If you bring historically excluded groups into an environment that’s not inclusive, they are only surviving it.”

Rigorous human-centered methodologies **must** prioritize the spectrum of lived experiences and expertise, are necessary to foster robust and collaborative environments, policies, and services.

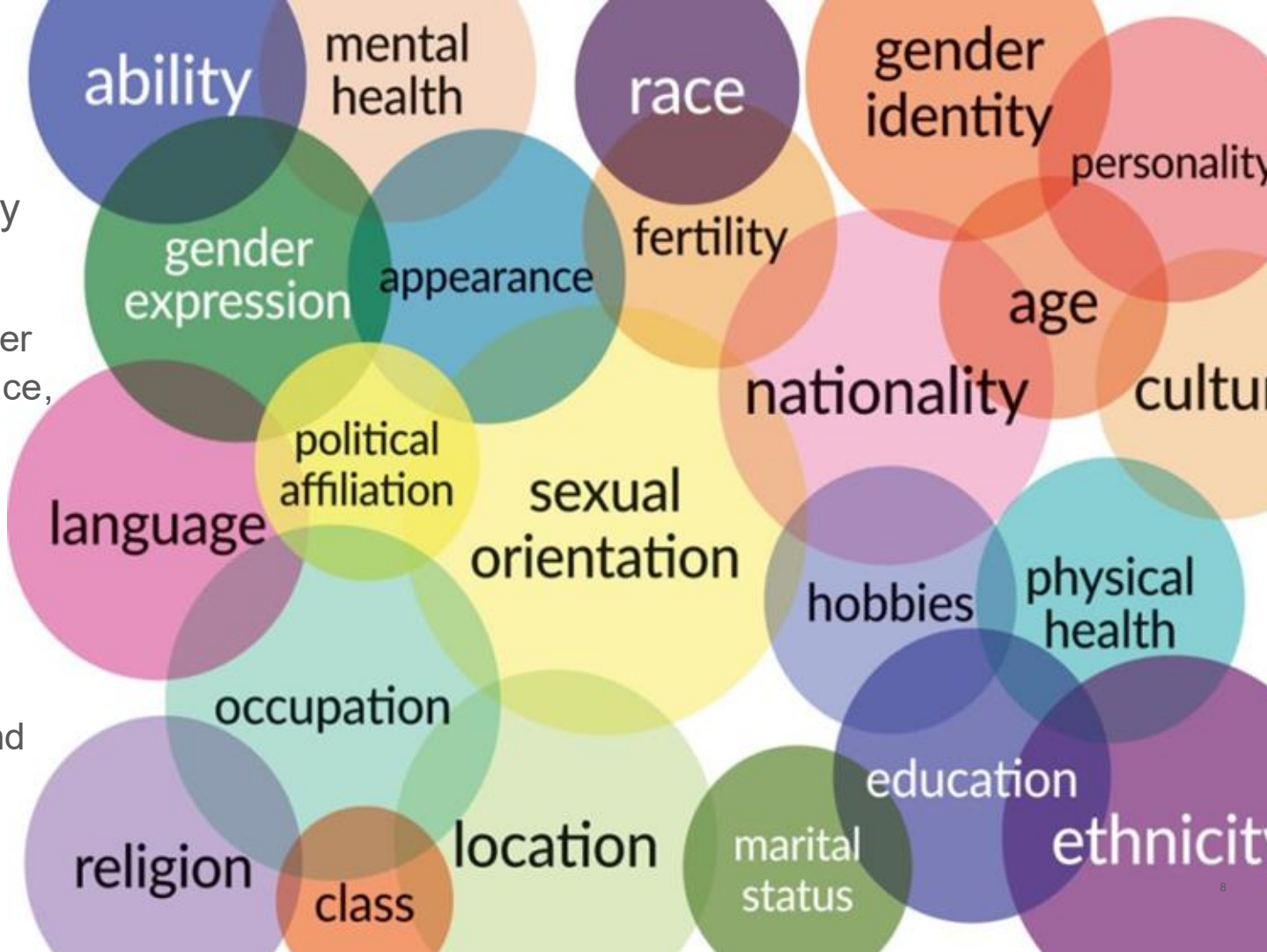
Intersectionality

One’s lived experience is greater than the sum of their identities.



Intersectionality may compound barriers

According to the Center for Intersectional Justice, intersectional identity characteristics may “intersect to create unique dynamics and effects” and may compound barriers to accessing services and impact outcomes.



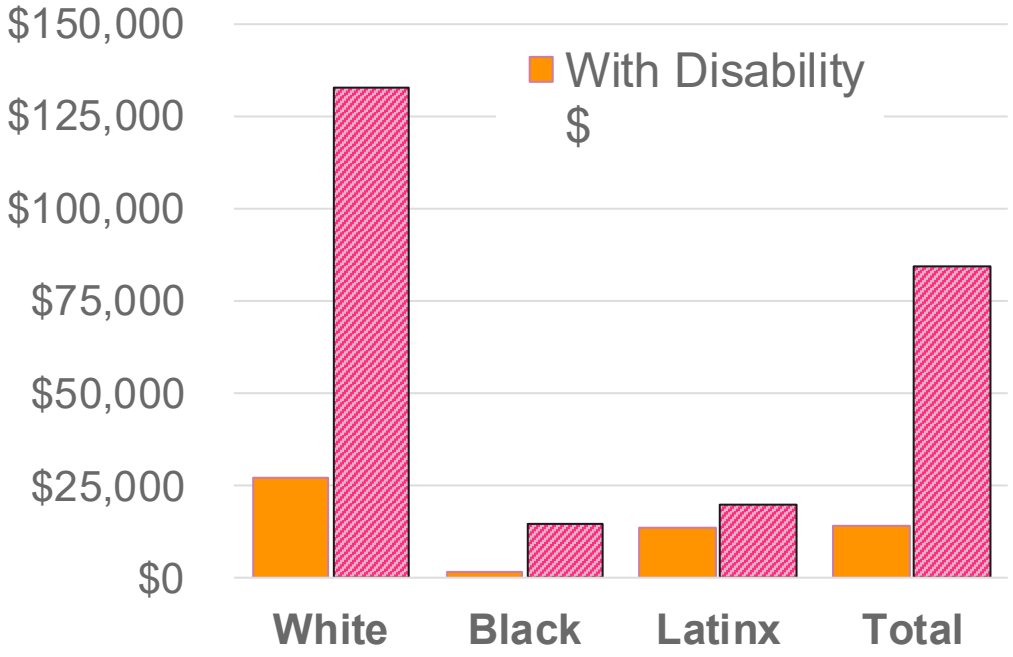
Matrix of Domination / Oppression

Social Identity	Privileged Social	Border Social	Targeted Social	-ism
Race	White People	Biracial People (White/Latinx, Black, Asian)	Asian, Black, Latinx, Native People	Racism
Sex	Bio Men	Transsexual, Intersex	Bio Women	Sexism
Gender	Gender-conforming Bio Men & Women	Gender Ambiguous Bio Men & Women	Transgender, Genderqueer, Intersex	Transgender Oppression
Sexual Orientation	Heterosexual	Bisexual	Lesbians, Gay Men	Heterosexism
Class	Rich, Upper Class	Middle Class	Working Class, Poor	Classism
Ability/Disability	Temporarily Able-bodied	Temporary Disabilities	People with Disabilities	Ableism
Religion	Protestants	Roman Catholic (historically)	Jews, Muslims, Hindu	Religious Oppression
Age	Adults	Young Adults	Elders, Young	Ageism/Adultism

Source: Teaching for Diversity and Social Justice, Second Edition *from* Routledge 2007 *and* https://www.deanza.edu/faculty/lewisjulie/MOD.pdf

Compounding effects of race & disability on net worth

Household net worth by race / ethnicity and disability status of working-age householder, 2016



There are significant disparities in household net worth by disability status and race.

- Black households with a disability have an average net worth of only $1,282.
- Latinx households with a disability have a $13,340 average net worth.
- In comparison, white households with a disability have a $27,100 average net worth.

Source: National Disability Institute Analysis of U.S. Census Bureau. 2019. Survey of Income & Program Participation Microdata, 2014 Panel, Wave 4.

Research on the compounding barriers of intersectionality

- **Complex Dynamics: Disability, Assistive Technology, and the LGBTQIA+ Community Center Experience in the United States**
  Crawford, Kirk; Spiel, Katta; Hamidi, Foad. October 2023. ASSETS ‘23.
  Study on the experience and barriers faced by LGBTQIA+ people with disabilities in community centers, identified “barriers to accessing and using AT, including social stigma, cost, and limitations in AT design.”
- **Working at the Intersection of Race, Disability, and Accessibility**
  Harrington, Christina; Moharana, Sanika; Desai, Aashaka; Spencer Ross, Anne; Lewis, Aaleyah; Mankoff, Jennifer. October 2023. ASSETS ’23.
  Examination of the intersection of race and disability to improve accessibility research, highlighting more work to be done especially continued engagement with representation from diverse communities.
- **Introduction to Intersectional Qualitative Research**
  Esposito, Jennifer; Evans-Winters, Venus. 2022. Sage Publications.
  Detailed book on the revolution of qualitative inquiry, including methodologies, analysis, conversational approaches.

We need rigorous, human-centered research methods to ensure organizations, both academic and work, consider the spectrum of humanity's lived experiences and expertise.

- When it comes to the public sector, multiple perspectives must be considered for policy effectiveness.
- If only quantitative data is considered, policy will fail as shown by numerous initiatives that failed to consider qualitative data.

New human-centered qualitative research methods…

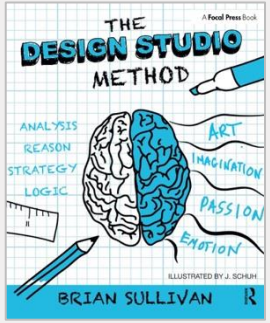
…can reduce bias, highlight all voices, and demonstrate inclusive approaches to meet mandates of the 2022 CHIPS & Science Act to measure “ability to attract and retain a diverse and nontraditional student [and staff] population in the fields of science, technology, engineering, and mathematics.”

To meet this act’s mandate, we recruited diverse perspectives, especially historically excluded populations, noted intersectional data patterns, and used automated verbatim transcription to reduce biases from notetaker interpretation.

Method Development

Background

From more than 25 years of experience across academia, government, private sector industry working with well-known researchers **Tom Tullis** and **Brian Sullivan** (author of The Design Studio Method), earning a certificate in problem space research with **Indi Young**, learning from **Creative Reaction Lab** and **HmntyCntrd**, reading numerous books…



CREATIVE REACTION LAB

Indi Young

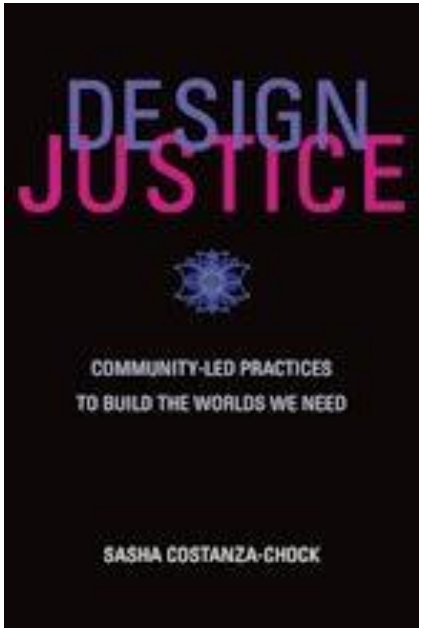
- Author of books, such as *Mental Models*, *Practical Empathy*, and *Time to Listen*.
- Offers an advanced training series and certification.
- Listening Sessions method incorporates a purpose-driven approach to develop “thinking styles.” It was developed with a private sector perspective and doesn’t incorporate facts and opinions.

Design Justice

Framework for analysis of how design distributes benefits and burdens between various groups of people.

Design justice focuses explicitly on the ways that design reproduces and/or challenges the matrix of domination (white supremacy, heteropatriarchy, capitalism, ableism, settler colonialism, and other forms of structural inequality).

Design justice is also a growing community of practice that aims to ensure a more equitable distribution of design's benefits and burdens; meaningful participation in design decisions; and recognition of community-based, Indigenous, and diasporic design traditions, knowledge, and practices.



Combining Design Justice and Indi Young's Listening Sessions, in an adapted variation…

- Using the Design Justice framework's recruiting **gaps analysis** (example follows) with Indi Young's Listening Sessions approach, adapted for the public sector to **include facts and opinions**, develops a rigorous, human-centered approach to qualitative analysis, applied to the CHIPS & Science Act requirement to inform criteria for broadening participation in institutions and organizations.
- It builds on established practices such as environmental and literature scans, while incorporating and building upon these innovative techniques.
- The method prioritizes inclusivity, minimizes researcher bias, and centers participants' own words to generate actionable insights for public policy design and implementation.

The Method

The Steps

A. Strategic, thoughtful recruiting informed by gaps analysis

B. Semi-structured interviews follow participant's lead

C. Verbatim transcript to capture participant's own words and avoid notetaker bias

D. Analysis that prioritizes participant mental models and purpose, retains the person's words

E. Synthesis into mental model diagrams (MMDs), a clear structure to cultivate cognitive empathy

F. Opportunity mapping activity to make the mental models actionable and align policymakers

Table 1. Example of Gaps Analysis — Recruiting Sampling Requirement

Sampling Requirement	Requested	Participants
Total !	12	12
Relevant experience	100%	100%
No college degree !	1	1
Women	50%	60%
Men !	50%	33%
Other gender identity !	10%	0%
Not white !	50%	90%
< 35 years old !	2	1
> 55 years old !	2	3
Disability – vision !	2	1
Disability – hearing	2	2
Disability – mobility !	2	1
Disability – cognitive, learning !	2	0
Disability – Post Traumatic Stress !	2	1
Disability – Traumatic Brain Injury !	2	1

The red exclamation mark (**!**) indicates a priority demographic based on previous recruiting gaps in lived experience.

Note: During research planning, establish recruiting goals for various demographics that are relevant for the research goal, seeking a variety of lived experiences.

- Who's involved?
- Who benefits?
- Who's harmed?

Record the data on the final participants. This gaps analysis informs future research efforts to address the disparities.

Table 2. Example of Category Analysis — Demographics of Participants

Demographic category	P01	P02	P03	P04	…
Disability – vision	-	-	-	X	…
Disability – hearing	-	X	X	-	…
…	…	…	…	…	…
No college degree	-	X	-	X	…
< 35 years old	-	-	-	X	…
> 55 years old	X	X	-	-	…

Note: Document demographic characteristics of each participant. Data is collected during recruiting in a screening survey. Individual participants may identify in multiple categories representing their intersectional identity.

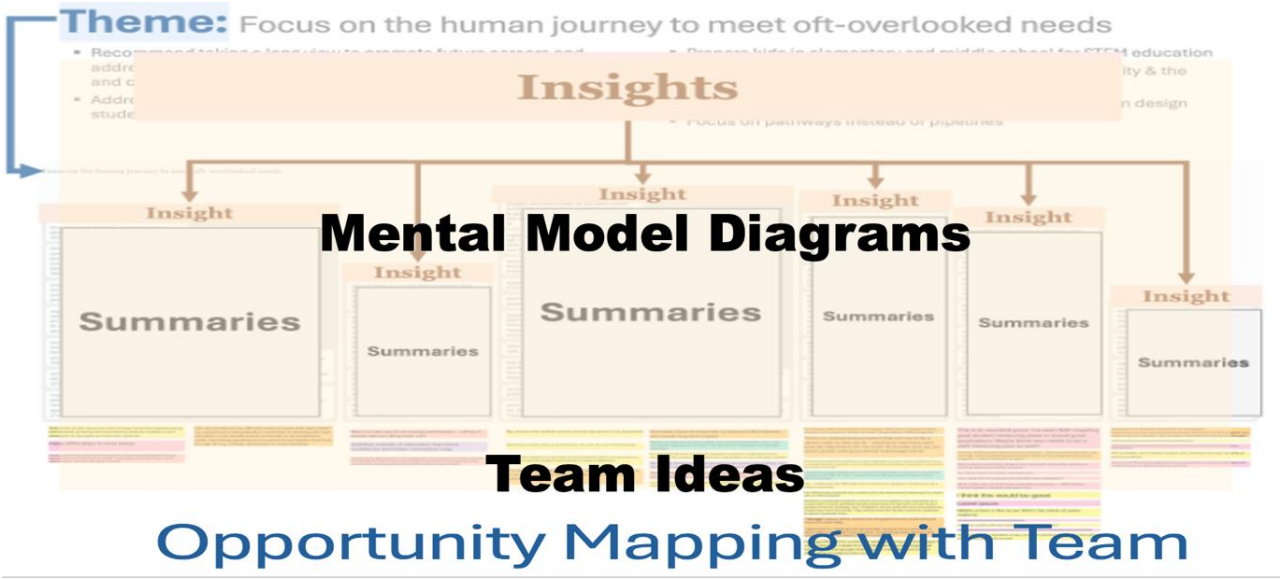
Figure 1. Specialized Template

This spreadsheet is used with Indi's Mental Model Dataset Visualizer to generate mental model diagrams, an organized representation of the themes, insights, and summaries.

	Theme	Insight	Summary	ID	Responses	Inner thinking / Reaction / Guiding Principle / Fact / Opinion / Story
1						
2	The theme that appears from a grouping of insights					
3		The insight that appears from a grouping of summaries				
4			active first person verb summary from participant's response with detail from their response. Each summary should be sentence that works with I + active first person verb + details.	P##	Participant's verbatim response, captured by automated transcription, broken into individual rows of separate thoughts / insights	What kind of insight is it? Inner thinking, reaction/feeling, guiding principle, fact, opinion, or story.
5			felt attracted to AI in the beginning because there's always something new, another opportunity to find creative solutions to new problems	P01	[Moderator: How do you roll with the changes as they come in the tech field?] Umm, that's part of what attracted me to AI in the beginning. It's, umm, there's always something new. There's always, you know, another opportunity to, umm, you know, find creative solutions to the new problems. I mean it's what keeps its interesting for me.	Reaction
6			ensure that AI isn't a space that is gatekept because AI requires diverse perspectives, passions, and interests as we introduce that technology into society	P03	I think AI is a really important area where. A lot of different, diverse perspectives in. Majors in topics of study and passions and interests and things like that are really valuable in that space. And I just wanna	guiding principle
7			find those teachable moments don't scale at a 10k person company, where people are resistant to ethics & bias stuff for whatever reasons	P02	So in some cases it was just finding, you know, finding those kinds of things. But that's a very one on one. It doesn't really scale when you're talking about trying to like. Each you know ethics and bias	inner thinking
8			...		...	...
9		The insight that appears from a grouping of summaries				
10			active first person verb summary from participant's response with detail from their response. Each summary should be sentence that works with I + active first person verb + details.	P##	Participant's verbatim response, captured by automated transcription, broken into individual rows of separate thoughts / insights	What kind of insight is it? Inner thinking, reaction/feeling, guiding principle, fact, opinion, or story.
11			dolor amet elit eiusmod tempor veniam nulla pariatur	P07	Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.	Story
12			...		...	...

Figure 2. Example Mental Model Diagram with Opportunity Mapping

- Participants' words are "combed" into insights and themes in specialized spreadsheet
- Generate mental model diagram (MMD)
- MMD are used in opportunity mapping with team to cultivate cognitive empathy



Results

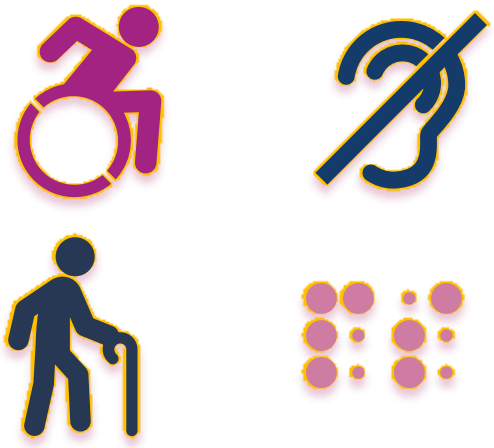
- Prioritized participants' own words to deliver actionable insights and elements that an inclusive culture or workplace needs for robust, resilient outcomes
- Provided evidence from people with disabilities that agencies used to evolve physical and digital services and policies
- Evolved Young's method for public sector where facts and opinions are necessary for policy success

Summary

Also, hiring that values lived experience is key to improving the systems that currently add barriers.

How do we do that legally, not discriminating?

- Applying principles of Design Justice in an environment resistant to concepts other than “meritocracy” can be challenging
- Focusing on serving all people is key, for example, noting one in four people have a disability
- Ensuring research strategy accommodates the inclusion of a broad range of perspectives will serve engineering and organization goals



Thank you!

ツ

Follow Us

linkedin.com/in/stricklandjen